LOCAL PERMUTATION POLYNOMIALS OF MAXIMUM DEGREE WITH COEFFICIENTS IN THE PRIME FIELD

KNOWLEDGE SENSE

Abstract. A polynomial in n variables over $\mathbb{F}_q$ is a local permutation polynomial if it permutes $\mathbb{F}_q$ in each variable when the other variables are fixed. Such a polynomial, reduced so that its degree in each variable is less than q, has total degree at most $n(q-2)$. We show that for every prime power $q = p^r > 3$ and every $n \geq 1$ this bound is attained by a local permutation polynomial whose coefficients lie in the prime field $\mathbb{F}_p$. The new ingredients are an exchange of two Frobenius orbits, which handles the case $p \geq 5$, $r \geq 2$, $p \nmid r$, and an exact computation of the top coefficient of a recursively defined family in characteristic 3. The same reduction shows that this recursive family, conjectured to have maximum degree in every odd characteristic, fails to do so for $(q, n) = (7, 10)$ and $(19, 8)$.

1. Introduction

Let $q = p^r$ be a power of a prime p. Every function $\mathbb{F}_q^n \to \mathbb{F}_q$ is represented by a unique polynomial $f \in \mathbb{F}_q[X_1, \ldots, X_n]$ with $\deg_{X_i} f < q$ for all i; we call such polynomials *reduced* and identify them with the functions they define. A reduced polynomial f is a *local permutation polynomial* (LPP) over $\mathbb{F}_q$ if, for each i and each choice of $c_j \in \mathbb{F}_q$ ($j \neq i$), the polynomial

$$f(c_1, \ldots, c_{i-1}, X_i, c_{i+1}, \ldots, c_n) \in \mathbb{F}_q[X_i]$$

permutes $\mathbb{F}_q$. In two variables these are exactly the polynomials describing Latin squares, and in general they describe Latin hypercubes. Local permutation polynomials over prime fields were studied by Mullen [6].

The total degree of a reduced LPP in n variables is at most $n(q-2)$ when $q > 2$; see [2, Proposition 1] and Lemma 2.1(d) below. For $n = 2$, Diestelkamp, Hartke and Kenney [1] showed that this bound is sharp. Gutierrez and Jiménez Urroz [3] constructed LPPs of degree $n(q-2)$ for every $q > 3$ and every $n \geq 1$. In characteristic 2, and whenever there is an integer $1 < b < p-1$ with $\gcd(b, q-1) = 1$, their polynomials have coefficients in $\mathbb{F}_p$ [3, Theorems 4 and 5]; in the remaining cases the construction uses coefficients outside $\mathbb{F}_p$ [3, Theorem 6]. They asked for LPPs of maximum degree defined over $\mathbb{F}_p$ for every q [3, Section 6]. This question was also considered in [4]. By [4, Theorem 1], for $p \geq 5$ such polynomials exist for every $q = p^r$ when $n < p-1$, and also for $n = p-1$ when $r \geq 2$; Theorem 2 there settles the case of odd p with $p \mid r$ for every n. In addition, for odd p the polynomials f_2 and f_3 of (1) below have maximum degree [3, Theorems 7 and 8].

Our main result answers this question affirmatively for every $q > 3$ and every n.

Theorem 1.1. *Let $q = p^r > 3$ and $n \geq 1$. There is a reduced local permutation polynomial $f \in \mathbb{F}_p[X_1, \ldots, X_n]$ over $\mathbb{F}_q$ with $\deg f = n(q-2)$.*

For $n = 1$ one may take X^{q-2}, which permutes $\mathbb{F}_q$ since $\gcd(q-2, q-1) = 1$. For arbitrary n, the results of [3, 4] leave open two cases:

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(A) $p \geq 5$, $r \geq 2$ and $p \nmid r$, when no integer $1 < b < p-1$ is coprime to $q-1$ (for example $q = 25$);
(B) $p = 3$ and $3 \nmid r$.

We handle case (A) with a switching lemma; the resulting construction covers every n and all of $p \geq 5$, $r \geq 2$, $p \nmid r$, including parameters already covered in [3, 4]. We handle case (B) by analyzing a recursive family introduced in [3].

The switching lemma (Lemma 3.1) states that if h is a permutation of $\mathbb{F}_q$ and $T \subseteq \mathbb{F}_q$ is a set with $h(T)$ invariant under $x \mapsto x+1$, then

$$\sum_{i=1}^{n} h(X_i) + \prod_{i=1}^{n} \mathbf{1}_T(X_i)$$

is an LPP, and it has maximum degree as soon as the elements of T have nonzero sum. Its coefficients lie in $\mathbb{F}_p$ when h and T are compatible with the Frobenius map. The cases $p = 2$ and $p \mid r$ are covered by this lemma with suitable choices of h and T (Remark 3.2 and Section 6). For $p \geq 5$, $r \geq 2$ and $p \nmid r$ we obtain suitable data by exchanging two Frobenius orbits of elements of degree r (Theorem 4.3).

For case (B) we study the recursively defined LPPs

$$f_1 = X_1, \qquad f_n = t\big(f_{n-1}^{q-2} + X_n^{q-2}\big) \quad (n \geq 2), \tag{1}$$

reduced in each variable, where $t(X) = X + \sum_{j=0}^{q-2} X^j$ is the transposition of 0 and 1. These polynomials lie in $\mathbb{F}_p[X_1, \ldots, X_n]$. Gutierrez and Jiménez Urroz proved that $\deg f_3 = 3(q-2)$ for odd p and conjectured that $\deg f_n = n(q-2)$ whenever $q = p^r > 3$ and p is odd [3, Theorem 8 and Conjecture 1]; since $f_1 = X_1$, the conjecture concerns $n \geq 2$. We express the coefficient of $X_1^{q-2} \cdots X_n^{q-2}$ in f_n in terms of the iterates of an explicit linear operator on univariate polynomials (Proposition 5.3). In characteristic 3 the orbit of U under this operator is periodic of period 2 from the first step on, which gives the following.

Theorem 1.2. *Let $q = 3^r$ with $r \geq 2$. For every $n \geq 2$ the coefficient of $X_1^{q-2} \cdots X_n^{q-2}$ in f_n equals 2. In particular* $\deg f_n = n(q-2)$.

Outside characteristic 3 the conjecture can fail.

Theorem 1.3. *For $q = 7$ one has* $\deg f_{10} < 10(q-2)$*, and for $q = 19$ one has* $\deg f_8 < 8(q-2)$.

The proof of Theorem 1.1 uses (1) only for $p = 3$, so Theorem 1.3 has no bearing on it.

The paper is organized as follows. Section 2 collects standard facts. Section 3 proves the switching lemma, and Section 4 constructs the orbit exchange. Section 5 analyzes the family (1) and proves Theorem 1.2. Theorem 1.1 is proved in Section 6, and Theorem 1.3 in Section 7.

2. Preliminaries

Throughout, $q = p^r$ and σ denotes the Frobenius map $x \mapsto x^p$ of $\mathbb{F}_q$. For a polynomial P we write $\mathrm{Red}(P)$ for the reduced polynomial defining the same function on $\mathbb{F}_q^n$; in one variable U, Red replaces U^e ($e \geq 1$) by $U^{e'}$, where $1 \leq e' \leq q-1$ and $e' \equiv e \pmod{q-1}$. We use the convention $0^0 = 1$. For a subset $T \subseteq \mathbb{F}_q$, $\mathbf{1}_T$ denotes its indicator function, with values in $\{0,1\} \subseteq \mathbb{F}_p$.

The following facts are standard (see [5] for (a)–(c) and [2, Proposition 1] for (d)); we include short proofs for completeness.

Lemma 2.1. *Let $q > 2$.*

(a) *Let $\varphi\colon \mathbb{F}_q^n \to \mathbb{F}_q$ and let P be its reduced polynomial. For $1 \le e_1, \dots, e_n \le q-1$, the coefficient of $X_1^{e_1} \cdots X_n^{e_n}$ in P is*

$$(-1)^n \sum_{x \in \mathbb{F}_q^n} \varphi(x)\, x_1^{q-1-e_1} \cdots x_n^{q-1-e_n}.$$

(b) *If $\varphi\colon \mathbb{F}_q \to \mathbb{F}_q$ satisfies $\varphi \circ \sigma = \sigma \circ \varphi$, then its reduced polynomial lies in $\mathbb{F}_p[X]$.*

(c) *The reduced polynomial of a permutation of $\mathbb{F}_q$ has degree at most $q-2$.*

(d) *If f is a reduced LPP in n variables, then $\deg_{X_i} f \le q-2$ for every i. Hence $\deg f \le n(q-2)$, with equality if and only if the coefficient of $X_1^{q-2} \cdots X_n^{q-2}$ in f is nonzero.*

Proof. (a) In $\mathbb{F}_p[X]$ we have $(1+X)^{q-1} = \sum_{j=0}^{q-1} (-X)^j$, since $(1+X)\sum_{j=0}^{q-1}(-X)^j = 1 + X^q = (1+X)^q$. Hence $(X-a)^{q-1} = \sum_{j=0}^{q-1}(-1)^j(-a)^{q-1-j}X^j = \sum_{j=0}^{q-1} a^{q-1-j}X^j$, because $q-1$ is even when p is odd (and signs are irrelevant when $p=2$). Since $1-(X-a)^{q-1}$ is the indicator function of $\{a\}$, the reduced polynomial of φ is

$$P = \sum_{x \in \mathbb{F}_q^n} \varphi(x) \prod_{i=1}^{n} \bigl(1 - (X_i - x_i)^{q-1}\bigr),$$

and the coefficient of $X_i^{e_i}$ with $e_i \ge 1$ in the i-th factor is $-x_i^{q-1-e_i}$.

(b) Write $P = \sum_j c_j X^j$ and $P^\sigma = \sum_j c_j^p X^j$. For $a \in \mathbb{F}_q$ we have $P^\sigma(a^p) = P(a)^p = \varphi(a)^p = \varphi(a^p)$. As σ is bijective, P^σ and P define the same function, and both are reduced, so $c_j^p = c_j$ for all j.

(c) By (a), the coefficient of X^{q-1} is $-\sum_{a \in \mathbb{F}_q} \varphi(a) = -\sum_{b \in \mathbb{F}_q} b = 0$.

(d) Write $f = \sum_{j=0}^{q-1} A_j X_i^j$ with A_j reduced in the remaining variables. For every specialization c of these variables, $\sum_j A_j(c) X_i^j$ is the reduced polynomial of a permutation, so $A_{q-1}(c) = 0$ by (c). Thus $A_{q-1} = 0$. The monomial $X_1^{q-2} \cdots X_n^{q-2}$ is the only monomial of degree $n(q-2)$ allowed by these bounds. □

We also use the following elementary closure properties [3, Theorem 1]. If f and g are LPPs in disjoint sets of variables, then $f+g$ is an LPP. If f is an LPP and $\pi \in \mathbb{F}_q[X]$ permutes $\mathbb{F}_q$, then $\pi(f)$ is an LPP.

3. A switching lemma

Lemma 3.1. *Let $q > 2$, let h be a permutation of $\mathbb{F}_q$, and let $S, T \subseteq \mathbb{F}_q$. Suppose that $h \circ \sigma = \sigma \circ h$, that $\sigma(T) = T$, and that*

(i) *$S + 1 = S$;*
(ii) *$h(T) = S$;*
(iii) *$s_T := \sum_{a \in T} a \neq 0$.*

Let g be the reduced polynomial of $\mathbf{1}_T$, and identify h with its reduced polynomial. Then $g, h \in \mathbb{F}_p[X]$, $\deg g = q-2$ and $\deg h \le q-2$. Moreover, for every $n \ge 2$ the polynomial

$$F_n = \sum_{i=1}^{n} h(X_i) + \prod_{i=1}^{n} g(X_i) \in \mathbb{F}_p[X_1, \dots, X_n]$$

is a reduced LPP over $\mathbb{F}_q$ of degree $n(q-2)$.

Proof. Since T is σ-stable, $\mathbf{1}_T$ commutes with σ, so $g \in \mathbb{F}_p[X]$ by Lemma 2.1(b); similarly $h \in \mathbb{F}_p[X]$, and $\deg h \le q-2$ by Lemma 2.1(c). By Lemma 2.1(a),

$$g(X) = \sum_{a \in T} \bigl(1 - (X-a)^{q-1}\bigr) = -|T|\, X^{q-1} - s_T\, X^{q-2} + (\text{lower terms}).$$

By (i), S is a union of cosets of $\mathbb{F}_p$, so $|T| = |S| \equiv 0 \pmod p$ by (ii). Together with (iii), this gives $\deg g = q-2$ with leading coefficient $-s_T$.

Each variable occurs in F_n with degree at most $q-2$, so F_n is reduced. The terms $h(X_i)$ have degree at most $q-2 < n(q-2)$, so the coefficient of $X_1^{q-2} \cdots X_n^{q-2}$ in F_n is $(-s_T)^n \neq 0$ and $\deg F_n = n(q-2)$.

It remains to show that F_n is an LPP. By symmetry it suffices to consider the variable X_1. Fix $c_2, \dots, c_n \in \mathbb{F}_q$ and put $C = \sum_{i \ge 2} h(c_i)$. If some $c_i \notin T$, the specialization is $h(X_1) + C$, a permutation. Otherwise it is $\varphi(x) = h(x) + \mathbf{1}_T(x) + C$. By (ii) and (i), φ maps T bijectively onto $S + 1 + C = S + C$, and it maps $\mathbb{F}_q \setminus T$ bijectively onto $(\mathbb{F}_q \setminus S) + C$. Hence φ is a permutation. □

Remark 3.2. Let $p = 2$ and $q > 2$, and take $h(X) = X$ and $S = T = \mathbb{F}_q \setminus \{0, 1\}$. All hypotheses of Lemma 3.1 hold; in particular $s_T = \sum_{a \in \mathbb{F}_q} a - 0 - 1 = 1$. As a function, $\sum_{i=1}^{q-2} X^i$ equals $\mathbf{1}_T$, so F_n is the polynomial of [3, Theorem 4].

4. Exchanging Frobenius orbits

In this section p is odd, $r \ge 2$ and $p \nmid r$. Let $D \subseteq \mathbb{F}_q$ be the set of elements x with $\mathbb{F}_p(x) = \mathbb{F}_q$, and let Tr be the trace from $\mathbb{F}_q$ to $\mathbb{F}_p$. Translations by elements of $\mathbb{F}_p$ commute with σ, so the group $G = \langle \sigma \rangle \times \mathbb{F}_p$, of order pr, acts on D by $(\sigma^i, c) \cdot x = x^{p^i} + c$.

Lemma 4.1. *The action of G on D is free.*

Proof. Suppose $x^{p^i} + c = x$ with $x \in D$, $0 \le i < r$ and $c \in \mathbb{F}_p$. Applying σ^i repeatedly gives $x^{p^{ki}} = x - kc$ for all $k \ge 1$. Take $k = r/\gcd(i, r)$; then $r \mid ki$, so $x^{p^{ki}} = x$ and $kc = 0$ in $\mathbb{F}_p$. Since $k \mid r$ and $p \nmid r$, k is invertible in $\mathbb{F}_p$, so $c = 0$. Then $x^{p^i} = x$, and since x has degree r over $\mathbb{F}_p$, $i = 0$. □

Lemma 4.2. *If $p \ge 5$ and $r \ge 2$, then $|D| > pr$.*

Proof. Every element of $\mathbb{F}_q \setminus D$ lies in a subfield $\mathbb{F}_{p^d}$ with $d \mid r$, $d < r$. Therefore

$$|\mathbb{F}_q \setminus D| \le \sum_{d=1}^{r-1} p^d < \frac{p^r}{p-1} \le \frac{p^r}{4},$$

so $|D| > \frac{3}{4} p^r$. Moreover $p^{r-1} \ge 5^{r-1} \ge 2r$ for $r \ge 2$, by induction on r. Hence $|D| > \frac{3}{4} p \cdot 2r > pr$. □

Theorem 4.3. *Let $p \ge 5$, $r \ge 2$ and $p \nmid r$. There exist a permutation h of $\mathbb{F}_q$ and subsets $S, T \subseteq \mathbb{F}_q$ satisfying all hypotheses of Lemma 3.1. Consequently, for every $n \ge 2$ there is a reduced LPP over $\mathbb{F}_q$ in $\mathbb{F}_p[X_1, \dots, X_n]$ of degree $n(q-2)$.*

Proof. By Lemmas 4.1 and 4.2, D is a union of free G-orbits of size pr, and there are at least two of them. Choose $\alpha \in D$ and $\beta_0 \in D \setminus G\alpha$. Since $\mathrm{Tr}(\beta_0 + c) = \mathrm{Tr}(\beta_0) + rc$ and r is invertible in $\mathbb{F}_p$, there is $c \in \mathbb{F}_p$ with $\mathrm{Tr}(\beta_0 + c) \neq \mathrm{Tr}(\alpha)$. Put $\beta = \beta_0 + c$; then $\beta \in G\beta_0$, so $\beta \in D \setminus G\alpha$.

Let $O_\alpha = \{\alpha^{p^i} : 0 \le i < r\}$ and $O_\beta = \{\beta^{p^i} : 0 \le i < r\}$. These are σ-orbits of size r, and they are disjoint, because $O_\alpha \subseteq G\alpha$ and $O_\beta \subseteq G\beta$. Define h by $h(\alpha^{p^i}) = \beta^{p^i}$ and $h(\beta^{p^i}) = \alpha^{p^i}$ for $0 \le i < r$, and $h(x) = x$ for all other x. Then h is a permutation commuting with σ. Set

$$S = G\alpha = \{\alpha^{p^i} + c : 0 \le i < r,\ c \in \mathbb{F}_p\}, \qquad T = (S \setminus O_\alpha) \cup O_\beta.$$

Both sets are σ-stable, and $S + 1 = S$. Since $S \setminus O_\alpha$ is disjoint from $O_\alpha \cup O_\beta$, h fixes it pointwise, so $h(T) = (S \setminus O_\alpha) \cup O_\alpha = S$. Finally,

$$\sum_{a \in S} a = \sum_{i=0}^{r-1} \sum_{c \in \mathbb{F}_p} \big(\alpha^{p^i} + c\big) = p\,\mathrm{Tr}(\alpha) + r \sum_{c \in \mathbb{F}_p} c = 0,$$

because p is odd. Hence $s_T = \sum_{a \in S} a - \mathrm{Tr}(\alpha) + \mathrm{Tr}(\beta) = \mathrm{Tr}(\beta) - \mathrm{Tr}(\alpha) \neq 0$. The last assertion follows from Lemma 3.1. □

Remark 4.4. The hypotheses of Theorem 4.3 cannot simply be dropped. If $p \mid r$, the action need not be free: for $r = p$, a root ζ of $Z^p - Z - 1$ lies in D and is fixed by $(\sigma, -1)$, since $\zeta^p = \zeta + 1$. If $p = 3$ and $r = 2$, then $|D| = 6 = pr$, and there is only one orbit. Both situations are covered by other means in Section 6.

5. The recursive family

In this section $q = p^r > 3$ is arbitrary. The polynomial $t(X) = X + \sum_{j=0}^{q-2} X^j$ satisfies $t(0) = 1$, $t(1) = 0$ and $t(a) = a$ for $a \neq 0, 1$, because $\sum_{j=0}^{q-2} a^j = 0$ for $a \neq 0, 1$. Since X^{q-2} permutes $\mathbb{F}_q$, the closure properties of Section 2 show by induction that each f_n in (1) is an LPP in $\mathbb{F}_p[X_1, \dots, X_n]$.

For a reduced univariate polynomial H we put

$$\Delta(H) = H(1) - H(0), \qquad E(U) = \sum_{j=0}^{q-2} U^j, \qquad K_H = \mathrm{Red}(H \circ t),$$

and we define the $\mathbb{F}_p$-linear operator

$$\mathcal{T}(H) = \mathrm{Red}\big(K_H'(U^{q-2})\big).$$

Note that $H \circ t$ is reduced before it is differentiated; differentiation does not commute with Red.

Lemma 5.1. *Let H be reduced. Then $K_H = H + \Delta(H)\,E$. If $K_H = \sum_{j=0}^{q-1} k_j U^j$, then*

$$\mathcal{T}(H) = k_1 - \sum_{m=1}^{q-2} m\,k_{q-m}\,U^m.$$

Proof. As functions, $H \circ t - H$ takes the value $\Delta(H)$ at 0, the value $-\Delta(H)$ at 1, and 0 elsewhere. The function E takes the values 1 at 0, $q - 1 = -1$ at 1, and 0 elsewhere. Since $H + \Delta(H)E$ is reduced, it equals K_H.

Next, $K_H'(U^{q-2}) = \sum_{j=1}^{q-1} j k_j U^{(j-1)(q-2)}$. The term $j = 1$ is the constant k_1. For $2 \le j \le q - 1$ we have $(j-1)(q-2) \ge 1$ and $(j-1)(q-2) \equiv -(j-1) \equiv q - j \pmod{q-1}$ with $1 \le q - j \le q - 2$, so $\mathrm{Red}(U^{(j-1)(q-2)}) = U^{q-j}$. Writing $m = q - j$, the coefficient of U^m is $(q-m)k_{q-m} = -mk_{q-m}$. □

Lemma 5.2. *Let $n \ge 2$ and let H be a reduced univariate polynomial. The coefficient of X_n^{q-2} in $\mathrm{Red}(H(f_n))$, viewed as a reduced polynomial in $X_1, \dots, X_{n-1}$, equals $\mathrm{Red}\big(\mathcal{T}(H)(f_{n-1})\big)$.*

Proof. Write $K = K_H = \sum_j k_j U^j$, and set $V = f_{n-1}^{q-2}$ and $Y = X_n^{q-2}$. As functions, $H(f_n) = H(t(V+Y)) = K(V+Y)$. In $\mathbb{F}_p[V,Y]$ we have the Taylor expansion

$$K(V+Y) = \sum_{m=0}^{q-1} K^{[m]}(V)\, Y^m, \qquad K^{[m]}(V) = \sum_j \binom{j}{m} k_j V^{j-m},$$

with $K^{[1]} = K'$. Now $\operatorname{Red}(Y^m) = X_n^{e(m)}$, where $e(0) = 0$ and, for $1 \le m \le q-1$, $e(m) \in [1, q-1]$ with $e(m) \equiv m(q-2) \equiv -m \pmod{q-1}$. Thus $e(m) = q-2$ if and only if $m = 1$. Collecting terms gives

$$\operatorname{Red}(H(f_n)) = \sum_{m=0}^{q-1} \operatorname{Red}\big(K^{[m]}(V)\big) X_n^{e(m)},$$

and the coefficient of X_n^{q-2} is $\operatorname{Red}(K'(f_{n-1}^{q-2})) = \operatorname{Red}(\mathcal{T}(H)(f_{n-1}))$, since $\mathcal{T}(H)$ and $K'(U^{q-2})$ define the same function. □

Proposition 5.3. *For $n \ge 1$, the coefficient c_n of $X_1^{q-2} \cdots X_n^{q-2}$ in f_n equals the coefficient of U^{q-2} in $\mathcal{T}^{n-1}(U)$.*

Proof. We show by descending induction on k, $1 \le k \le n$, that the coefficient of $X_{k+1}^{q-2} \cdots X_n^{q-2}$ in f_n, as a reduced polynomial in $X_1, \dots, X_k$, equals $\operatorname{Red}(\mathcal{T}^{n-k}(U)(f_k))$. For $k = n$ this is trivial. If it holds for some $k \ge 2$, then taking the coefficient of X_k^{q-2} and applying Lemma 5.2 with n replaced by k and $H = \mathcal{T}^{n-k}(U)$ gives the statement for $k-1$. For $k = 1$ we obtain $\mathcal{T}^{n-1}(U)(X_1)$, and the claim follows. □

Remark 5.4. For odd p, Lemma 5.1 gives $\mathcal{T}(U) = 2 - \sum_{m=2}^{q-2} mU^m$, whose coefficient of U^{q-2} is $-(q-2) = 2 \neq 0$. Hence $\deg f_2 = 2(q-2)$, in agreement with [3, Theorem 7].

Proof of Theorem 1.2. Let $q = 3^r$ with $r \ge 2$. Throughout the proof we regard integers as elements of $\mathbb{F}_3$. Each residue class modulo 3 occurs $q/3$ times among $0, 1, \dots, q-1$, and $3 \mid q/3$; hence $\sum_{m=0}^{q-1} m = 0$ and $\sum_{m=0}^{q-1} m^2 = 0$ in $\mathbb{F}_3$. Since $q - 1 = -1$ in $\mathbb{F}_3$, it follows that

$$\sum_{m=2}^{q-2} m = 0 - 1 - (q-1) = 0, \qquad \sum_{m=2}^{q-2} m^2 = 0 - 1 - (q-1)^2 = 1. \tag{2}$$

Define

$$A = 2 - \sum_{m=2}^{q-2} mU^m, \qquad B = -\sum_{m=2}^{q-2} m^2 U^m.$$

We claim that $\mathcal{T}(U) = A$, $\mathcal{T}(A) = B$ and $\mathcal{T}(B) = A$. Each claim follows from Lemma 5.1.

$H = U$. Here $\Delta(U) = 1$, so $k_0 = 1$, $k_1 = 2$, $k_j = 1$ for $2 \le j \le q-2$, and $k_{q-1} = 0$. Hence $\mathcal{T}(U) = 2 - \sum_{m=2}^{q-2} mU^m = A$.

$H = A$. By (2), $\Delta(A) = -\sum_{m=2}^{q-2} m = 0$, so k_j is the coefficient of U^j in A: $k_1 = k_{q-1} = 0$ and $k_j = -j$ for $2 \le j \le q-2$. Hence $\mathcal{T}(A) = -\sum_{m=2}^{q-2} m \cdot \big(-(q-m)\big) U^m = -\sum_{m=2}^{q-2} m^2 U^m = B$.

$H = B$. By (2), $\Delta(B) = -1 = 2$, so $k_1 = 2$, $k_{q-1} = 0$ and $k_j = 2 - j^2$ for $2 \le j \le q-2$. Hence

$$\mathcal{T}(B) = 2 - \sum_{m=2}^{q-2} m\big(2 - (q-m)^2\big) U^m = 2 - \sum_{m=2}^{q-2} (2m - m^3) U^m = A,$$

since $m^3 = m$ in $\mathbb{F}_3$.

Consequently $\mathcal{T}^{n-1}(U) \in \{A, B\}$ for every $n \geq 2$. The coefficient of U^{q-2} is $-(q-2) = 2$ in A and $-(q-2)^2 = -4 = 2$ in B. By Proposition 5.3, $c_n = 2$, and $\deg f_n = n(q-2)$ by Lemma 2.1(d). □

6. Proof of Theorem 1.1

For $n = 1$ we take X^{q-2}. Let $n \geq 2$. We distinguish cases.

Case $p = 2$. Use Remark 3.2; this is [3, Theorem 4].

Case $p = 3$. Since $q > 3$, we have $r \geq 2$, and f_n has the required properties by Theorem 1.2.

Case $p \geq 5$, $r = 1$. Take $b = p - 2$. Then $1 < b < p - 1$ and $\gcd(b, p-1) = 1$, so [3, Theorem 5] gives the required polynomial.

Case $p \geq 5$, $p \mid r$. This case is [4, Theorem 2]. It also follows from Lemma 3.1, as we now indicate. Let ζ be a root of $P(Z) = Z^p - Z - 1$. Since P has no root in $\mathbb{F}_p$ and $\zeta^p = \zeta + 1$, the roots of P are $\zeta + c$ ($c \in \mathbb{F}_p$), they form one σ-orbit of length p, and $\zeta \in \mathbb{F}_{p^p} \subseteq \mathbb{F}_q$. Let $h(X) = X^{q-2}$, which acts as inversion on $\mathbb{F}_q^\times$ and commutes with σ. Put $S = \zeta + \mathbb{F}_p$ and $T = h(S) = \{(\zeta + c)^{-1} : c \in \mathbb{F}_p\}$; note that $0 \notin S$. Then T is σ-stable, $S + 1 = S$ and $h(T) = S$. From $P(Z) = \prod_{c \in \mathbb{F}_p}(Z - \zeta - c)$ we get

$$\sum_{c \in \mathbb{F}_p} \frac{1}{\zeta + c} = -\frac{P'(0)}{P(0)} = -1 \neq 0.$$

Thus all hypotheses of Lemma 3.1 hold, and F_n is the required polynomial.

Case $p \geq 5$, $r \geq 2$, $p \nmid r$. Apply Theorem 4.3.

These cases exhaust all $q = p^r > 3$, which proves Theorem 1.1. □

7. The recursive family in other characteristics

By Proposition 5.3 and Lemma 2.1(d), $\deg f_n = n(q-2)$ if and only if the coefficient of U^{q-2} in $\mathcal{T}^{n-1}(U)$ is nonzero. Lemma 5.1 describes $\mathcal{T}$ as an explicit linear map on $\mathbb{F}_p^q$, so these coefficients can be computed exactly for any given q and n. All computations below were carried out in exact arithmetic modulo p; a verification script in Python 3, using only the standard library, is available from the author on request.

Proof of Theorem 1.3. For $q = 7$, Table 1 lists $\mathcal{T}^k(U)$ for $0 \leq k \leq 9$; each row is obtained from the previous one by Lemma 5.1. The coefficient of U^6 is always 0 and is omitted. In particular

$$\mathcal{T}^9(U) = 6 + 4U^3 + 3U^4 \in \mathbb{F}_7[U],$$

whose coefficient of U^5 vanishes. Hence $c_{10} = 0$ and $\deg f_{10} < 50$.

For $q = 19$ the same iteration gives

$$\begin{aligned}\mathcal{T}^7(U) = 2 &+ U^3 + 16U^4 + 10U^5 + U^6 + 11U^7 + 4U^8 + 6U^9 + 13U^{10} \\ &+ 15U^{11} + 8U^{12} + 18U^{13} + 9U^{14} + 3U^{15} + 18U^{16} \in \mathbb{F}_{19}[U],\end{aligned}$$

with zero coefficient of U^{17}. Hence $c_8 = 0$ and $\deg f_8 < 136$. □

The values of c_n for these two fields are

$$\begin{aligned} q = 7: &\quad (c_1, \ldots, c_{10}) = (0, 2, 3, 3, 1, 6, 2, 3, 1, 0), \\ q = 19: &\quad (c_1, \ldots, c_8) = (0, 2, 15, 15, 8, 1, 17, 0). \end{aligned}$$

Thus, among $n \geq 2$, the values $n = 10$ and $n = 8$ are the smallest with $\deg f_n < n(q-2)$ for $q = 7$ and $q = 19$, respectively.

TABLE 1. The polynomials $\mathcal{T}^k(U) = \sum_m u_m U^m$ over $\mathbb{F}_7$, and $\Delta = \Delta(\mathcal{T}^k(U))$. The coefficient c_{k+1} of Proposition 5.3 is u_5.

k	u_0	u_1	u_2	u_3	u_4	u_5	Δ
0	0	1	0	0	0	0	1
1	2	0	5	4	3	2	0
2	0	0	3	5	5	3	2
3	2	0	4	0	0	3	0
4	0	0	1	0	0	1	2
5	2	0	1	1	6	6	0
6	0	0	2	3	3	2	3
7	3	0	4	3	4	3	0
8	0	0	1	2	2	1	6
9	6	0	0	4	3	0	0

Remark 7.1. For a prime $q = p$ there is an independent way to compute c_n which avoids polynomial reduction. By Lemma 2.1(a),

$$c_n = (-1)^n \sum_{x \in \mathbb{F}_p^n} f_n(x)\, x_1 \cdots x_n = (-1)^n \sum_{z \in \mathbb{F}_p} z\, W_n(z), \qquad W_n(z) = \sum_{f_n(x) = z} x_1 \cdots x_n.$$

Writing $\iota(x) = x^{p-2}$, the recursion (1) gives $W_1(z) = z$ and

$$W_n(z) = \sum_{\substack{y, x \in \mathbb{F}_p \\ t(\iota(y) + \iota(x)) = z}} W_{n-1}(y)\, x \qquad (n \geq 2).$$

For $p = 7$ this yields $(W_{10}(0), \ldots, W_{10}(6)) = (1, 1, 6, 4, 6, 4, 6)$, and $\sum_z z W_{10}(z) = 105 = 0$ in $\mathbb{F}_7$. For both $q = 7$ and $q = 19$, this computation reproduces the values of c_n listed above.

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